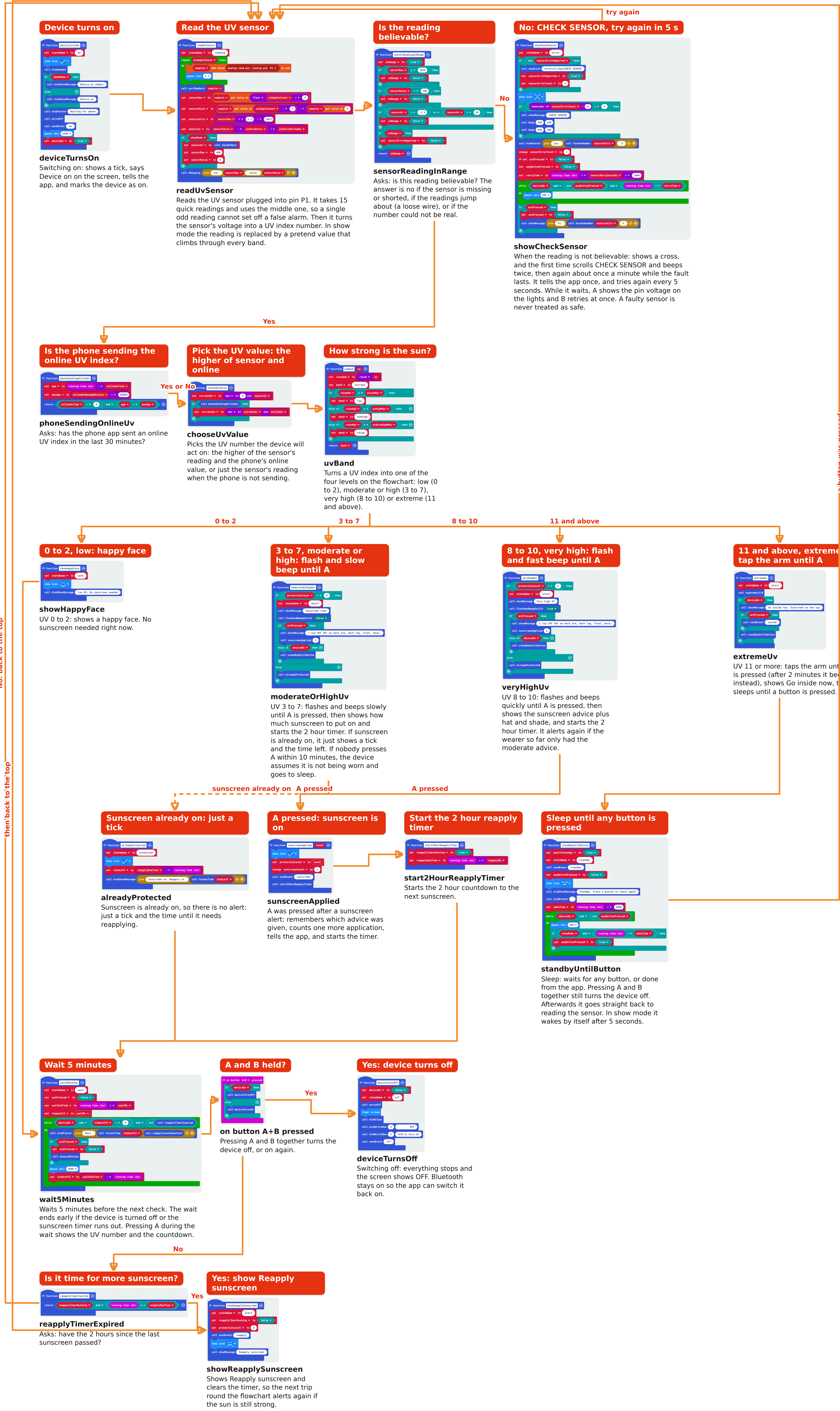
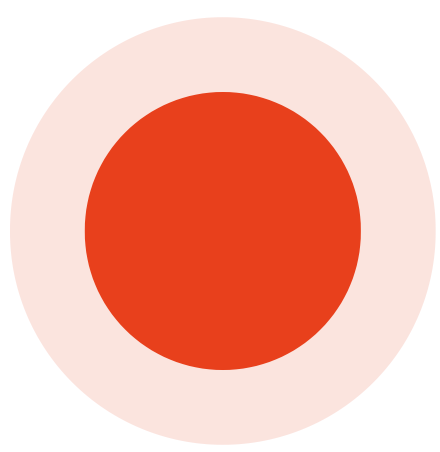


Sunburn device: the program on the wrist

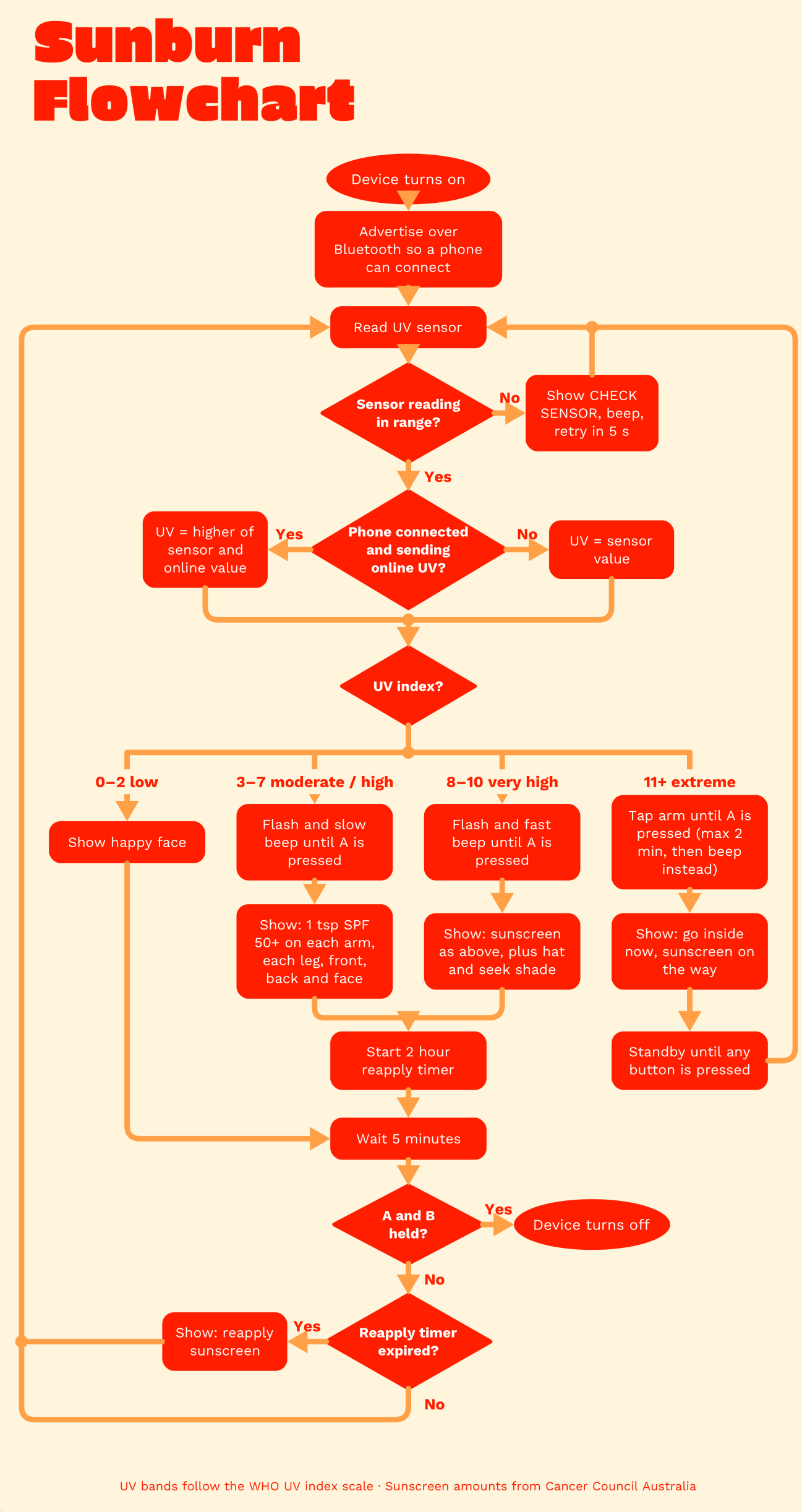
The Sunburn Flowchart, built from the real blocks of the program, with what each one does

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights and two buttons. It measures the sun's UV with a sensor, checks the online UV index for where you are through a phone, and then flashes, beeps or taps your arm until you put sunscreen on. Two hours later it reminds you to reapply. The program that does all this is built from 78 coloured blocks. Every one of them is on these sheets, with a note saying what it does.

Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block of code that does that job, exactly as it looks in the MakeCode editor, and the note says what it does. The forever block runs this plan round and round, all day.



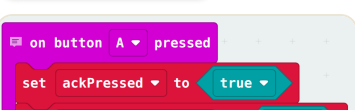
The Sunburn Flowchart



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

A: done

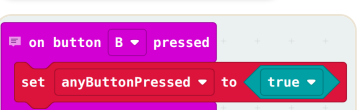


on button A pressed

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

Pressing A and B together turns the device off, or on again (see the flowchart).

B: wake up



on button B pressed

Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.

The whole program



All 78 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

What the block colours mean

Variables: remember a value	Text
Functions: a step with a name	Arrays: lists
Basic: the lights, pauses	Music: beeps
Input: buttons and time	Pins: sensor, motor, screen
Loops: repeat	Bluetooth: the phone
Logic: if, compare	Serial: the USB cable
Maths	Control: timing

Words used here

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 78 blocks, with its note, is in the booklet on the table.